

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – CASE STUDY

Course Code:	CSA0615	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO3: Articulation (BL4, BL5)	Assessment:	Assessment Tool 3 – Case Study
Weightage:	20% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO3 Statement:	Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication		
Student Name:	V.LASYA	Reg. No.:	192524186
Section / Batch:	B.TECH AI&DS	Date:	17-08-2026

Instructions to Students

- Answer both questions. Each question carries 50 Marks. Total: 100 Marks.
- Carefully analyze the given case study questions.
- Explain in detail with sample code if needed.
- Clearly identifies all key issues, in-depth analysis, Innovative, feasible solutions with strong justification

Question Type	Description	Questions	Total Marks
Case Study	Focus on Design Divide and Conquer algorithms: Merge Sort, Quick Sort, Binary Search and Matrix Multiplication	Q1 – Q2	100

SECTION A – CASE STUDY

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
DAA PROGRAMMING

Vandana Lasya
192524186RA

CO3-AT3- Case Study

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Language: Python C C++ Java

QUESTIONS

Q1
Divide and Conquer
50 marks

Q2
Divide and Conquer
50 marks

Q1 Divide and Conquer

50 marks

Case Study: Log File Analysis System (Binary Search + Sorting)
A system analyzes large log files by first sorting entries and then searching for specific events. Analyze how sorting (Merge/Quick Sort) and Binary Search can be combined effectively. Identify key issues such as processing time and data size. Apply divide and conquer techniques in both stages. Analyze overall time complexity. Design a solution and justify its efficiency.

Your Code

```
1 def merge_sort(a):
2     if len(a)<=1:
3         return a
4     m=len(a)//2
5     return sorted(merge_sort(a[:m]) +
6 merge_sort(a[m:]))
7 def binary_search(a,x):
8     l,r=0,len(a)-1
9     while l<=r:
10         m=(l+r)//2
11         if a[m]==x:
12             return m
13         if a[m]<x:
14             l=m+1
15         else:
16             r=m-1
17     return -1
18 n=int(input())
19 a=list(map(int,input().split()))
20 x=int(input())
21 a=merge_sort(a)
22 print(*a)
23 print(binary_search(a,x))
24
```

Run

Save

Submit

Input

```
5
40 10 30 20 50
30
```

Output

```
10 20 30 40 50
2
```

QUESTIONS

Q1
Divide and Conquer
50 marks

Q2
Divide and Conquer
50 marks

Q2 Divide and Conquer

50 marks

Case Study: Student Record Retrieval System (Binary Search)

A university database stores student details in sorted order and requires quick search operations. Analyze how Binary Search improves retrieval efficiency. Identify requirements such as sorted data and accuracy. Apply divide and conquer principles. Analyze time complexity. Design a solution and justify its effectiveness.

Your Code

```

1 n=int(input())
2 a=list(map(int,input().split()))
3 x=int(input())
4 low=0
5 high=n-1
6 while low<=high:
7     mid=(low+high)//2
8     if a[mid]==x:
9         print("Found")
10        break
11    elif a[mid]<x:
12        low=mid+1
13    else:
14        high=mid-1
15 else:
16    print("Not Found")

```

✓ Submitted

Input

```

5
10 20 30 40 50
30

```

Output

```

Found

```

Code of Conduct

I V.Lasya(192524186) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

V.Lasya

RUBRICS FOR CALCULATION

Case Study					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts

		analyze the case deeply			
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Mark Evaluation:

Case Study	Understanding of case (25)	Applications of concepts (25)	Analysis (20)	Solution Design (20)	Clarity (10)	Total Marks (Each - 50)
Q1						
Q2						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Name: _____ Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____